

# Chapter 110 — Overactive Bladder

## TERMINOLOGY AND DEFINITIONS

- **Overactive bladder (OAB) is a syndrome** of one or more storage symptoms driven by an excessively frequent or exceptionally strong desire to void.
- **International Continence Society (ICS) definition** and its **four component symptoms** — daytime frequency, nocturia, urgency, urgency urinary incontinence (UUI) — are set out in Table 110.1.
- **"Urge incontinence" was renamed "urgency incontinence"** — for consistency across terms and to tie the language to an abnormal *sensation*.
- **All components reduce health-related quality of life (QoL); urgency and UUI carry the strongest negative effect.** UUI is the **most bothersome lower urinary tract symptom (LUTS) in both sexes.**
- **Daytime frequency becomes bothersome at voiding every 2 hours or more often ( $\approx \geq 8$  daytime voids)** — but bother is individual and can be significant below eight voids.
- **Nocturia becomes bothersome at  $\geq 2$  episodes, markedly so at  $\geq 3$ .**
- **Isolated nocturia** (no other storage symptom) usually reflects **nocturnal polyuria or poor sleep, not lower urinary tract (LUT) dysfunction.**
- **OAB is a symptom diagnosis; detrusor overactivity (DO) is a urodynamic observation** — involuntary detrusor contractions during filling, spontaneous or provoked.
- **Not interchangeable:** OAB patients may have no DO; conversely DO may be asymptomatic or an **artefact of instrumentation.**

TABLE 110.1 STANDARDIZED TERMINOLOGY AND DEFINITIONS TO DESCRIBE LOWER URINARY TRACT DYSFUNCTION RELATED TO OVERACTIVE BLADDER

| Term                           | Definition                                                                                                                          |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Overactive bladder syndrome    | Urgency, usually with frequency and nocturia, $\pm$ UUI, in the absence of urinary tract infection (UTI) or other obvious pathology |
| Increased daytime frequency    | Patient considers they void too often by day; <b>upper limit of normal seven daytime voids</b>                                      |
| Nocturia                       | Interruption of sleep <b>one or more times</b> to micturate; <b>each void preceded and followed by sleep</b>                        |
| Urgency                        | Sudden, <b>compelling</b> desire to void that is <b>difficult to defer</b>                                                          |
| Urgency (urinary) incontinence | Involuntary loss of urine <b>associated with urgency</b>                                                                            |

Data from Abrams et al., 2002; Haylen et al., 2010 (ICS standardisation reports).

- **What the table is for:** fixing the standardised vocabulary on which every later threshold and trial endpoint depends.
- **Pattern:** every definition is a *complaint* — patient-reported, not measured.
- **Trap:** the nocturia definition counts one waking episode, yet bother only begins at  $\geq 2$ ; and a void that is not preceded *and* followed by sleep is not nocturia.

### Mixed Symptoms Incorporating Urinary Urgency

- **Mixed urinary incontinence (MUI):** involuntary leakage associated with **urgency and with effort, exertion, sneezing or coughing** — i.e. stress urinary incontinence (SUI) **plus UUI ("OAB wet")** in the same person.

- **SUI + "OAB dry" is not MUI** — no standard term exists; describe it (e.g. "SUI and urgency without UUI").
- **Urinary plus faecal incontinence = double (dual) incontinence**, not "mixed incontinence" — avoids confusion with MUI.
- **MUI generally means more severe leakage**; the relative contribution of each component is hard for patients to judge, especially with **insensible incontinence** (leakage the patient did not notice happening).

#### "OAB DRY" VERSUS "OAB WET"

|                      | OAB dry                                  | OAB wet                            |
|----------------------|------------------------------------------|------------------------------------|
| Defining feature     | Urgency <b>without</b> UUI               | Urgency <b>with</b> UUI            |
| Combined with SUI    | "SUI and urgency" — described, not named | <b>MUI</b>                         |
| Psychological burden | Lower                                    | <b>Higher</b>                      |
| DO on cystometry     | Less likely                              | More likely (almost always in men) |

#### Distinguishing Overactive Bladder From Bladder Pain Syndrome

- **Older definitions of urgency included fear of pain as well as fear of leakage**; pain is now **not** part of OAB but is **required** for bladder pain syndrome (BPS).
- **BPS: persistent or recurrent chronic pelvic pain, pressure or discomfort perceived to be related to the bladder, plus at least one other urinary symptom** (urgency or frequency).
- **Shared symptoms:** urgency, daytime frequency and nocturia; **UUI belongs to OAB, pain to BPS**.
- **Uncertain boundary:** up to about seven in ten women with bladder-centric pain also report urgency and/or UUI; both may be **variable expressions of shared afferent hypersensitivity, central sensitisation and somatisation**.

#### OAB VERSUS BPS — THE GOVERNING CONTRAST

| Feature                | OAB                                                                                            | BPS                                                         |
|------------------------|------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| Primary fear           | <b>Fear of leakage</b>                                                                         | <b>Fear of pain</b>                                         |
| Sensation with filling | <b>Phasic</b> waves of urgency that rise and recede until severe urgency forces voiding or UUI | <b>Steady, accelerating rise in pain</b> throughout filling |
| Voided volumes         | <b>Variable</b>                                                                                | <b>More consistent</b>                                      |
| Ability to defer       | Limited — leakage risk                                                                         | <b>Can defer</b> , at the cost of more pain                 |
| Site of sensation      | Perineum / base of penis, or vagina / urethra                                                  | <b>Suprapubic</b> ± perineal (urethral, vaginal, penile)    |
| UUI                    | Yes ("wet")                                                                                    | <b>No</b>                                                   |

#### KEY POINTS: OVERACTIVE BLADDER TERMINOLOGY

- **OAB = urgency-defined symptom syndrome; DO = urodynamic observation.**
- **MUI = UUI + SUI; fear of leakage (OAB) vs fear of pain (BPS), possibly both central sensitisation.**

**Concept** — OAB is defined by a sensation (urgency), not by a measurement: this is why DO, pain, SUI and polyuria each have to be separated from it explicitly, and why every "mixed" label depends on whether UUI is present.

## EPIDEMIOLOGY AND ECONOMICS

### Prevalence of Overactive Bladder and Related Lower Urinary Tract Symptoms

- **Prevalence of OAB in adults is reported between about 12% and 29%, UII between about 1.5% and 17%; the spread reflects differing case definitions and sampled populations, which limit direct comparison.**

**TABLE 110.2 STUDIES EXAMINING PREVALENCE DATA FOR OVERACTIVE BLADDER**

| Study                                                                    | Population                 | Method                                       | Definition of OAB                                                                                                               | Prevalence                                 |
|--------------------------------------------------------------------------|----------------------------|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| <b>NOBLE</b> (National Overactive Bladder Evaluation)                    | US adults $\geq 18$ y*     | Computer-assisted telephone interview (CATI) | <b><math>\geq 4</math> urgency episodes in preceding 4 weeks and <math>\geq 8</math> voids/day or bladder coping mechanisms</b> | OAB 16.5%; with UII 6.1%                   |
| <b>EpiLUTS</b> (Epidemiology of LUTS)                                    | Multinational, $\geq 40$ y | Internet survey                              | Urgency at least "sometimes" and/or UII                                                                                         | <b>OAB 28.7% (highest);</b> with UII 17.3% |
| <b>OAB-POLL</b> (Impact of OAB on Physical and Occupational Limitations) | US, 18–70 y*               | Internet survey                              | Urgency at least "sometimes" and/or UII                                                                                         | OAB 23.3%; with UII 12.6%                  |
| <b>NHANES</b> (National Health and Nutrition Examination Survey)         | US women $\geq 21$ y*      | Interview                                    | Any incontinence in past 12 months                                                                                              | UII 7.9%; with MUI 17%                     |
| <b>BACH</b> (Boston Area Community Health Survey)                        | Boston, 30–79 y*           | Interview                                    | <b>American Urological Association symptom index <math>\geq 8</math></b>                                                        | LUTS 18.7%                                 |
| <b>EPIC</b>                                                              | Multinational, $\geq 18$ y | CATI                                         | <b>ICS definitions</b>                                                                                                          | <b>OAB 11.8% (lowest);</b> with UII 1.5%   |
| Six-country European survey (Milsom)                                     | Multinational, $\geq 40$ y | Interview                                    | Any OAB symptom                                                                                                                 | Any OAB symptom 16.6%                      |

\*Selective sampling to allow generalisation to the US population.

- **What the table is for:** showing that **prevalence is a function of definition**.
- **Pattern:** the **loosest definition** (urgency "sometimes", internet survey, older population — EpiLUTS) gives the **highest** figure; the **strict ICS definition** (EPIC) gives the **lowest**.
- **Trap:** BACH measures **LUTS by symptom score**, not OAB; NHANES measures **incontinence**, not OAB — neither is an OAB prevalence.
- **Sex and age** (table below):
- In **NOBLE**, **overall OAB prevalence was similar in the two sexes, but UII was roughly three times commoner in women** — the sex difference lies chiefly in **wet OAB**.
- **EpiLUTS and OAB-POLL** found OAB in women roughly **double** that in men.
- **Beyond 75 years**, a third to two-fifths of both sexes report an OAB symptom, versus about one in ten or fewer before 55.
- **Race and ethnicity:** OAB and UII are **slightly higher in Black men and women** than in other groups across EpiLUTS, OAB-POLL and NHANES; **Asian women** had the lowest prevalence in EpiLUTS.

**SEX CONTRAST**

|                                | Women       | Men      |
|--------------------------------|-------------|----------|
| OAB prevalence                 | 1.5–2 × men | —        |
| UUI prevalence                 | 2–3 × men   | —        |
| Age at which OAB rises         | 40s         | 50s–60s  |
| Age at which UUI rises steeply | After 44    | After 64 |

**Incidence and Progression of Overactive Bladder**

- **OAB is dynamic** — symptoms wax and wane; longitudinal data are limited.
- **Incidence in women: about 3–8% per year.**
- **Progression from OAB dry to UUI: about 1% per year; remission of OAB: about 6% per year** — remission outpaces progression.

**Health Burden of Overactive Bladder**

- **OAB impairs daily activities, sleep, self-esteem, relationships and intimacy; UUI adds burden beyond dry OAB.** People with OAB report **less leisure-time physical activity.**
- **Work:** greater **impairment at work, more work time missed, higher unemployment,** and more absence from medical causes and disability.
- **Under-presentation:** fewer than half of people with OAB have ever discussed it with a clinician, although nearly all have seen one for other reasons — **a barrier specific to OAB, not to health care access in general.**
- Common reasons: **not thought important, not asked by the clinician, belief that it must simply be coped with or that nothing can be done.** Fewer than half of women report ever being asked.
- **Fewer than 6% of symptomatic people take medication.**
- **Anticholinergics are only modestly superior to placebo in aggregate, so persistence is poor:**
- **About 92% fail initial anticholinergic therapy within 2 years; most stop within 6 months; half never refill the first prescription.**
- Reasons for stopping, in order: **unmet expectations > side effects > cost.**

**Costs**

- **Direct and indirect costs** fall on patients and the health system; much of the **patient's cost is routine containment** (pads, diapers, laundry) and scales with incontinence severity.
- **Disparities:** a **greater containment-product burden in non-Hispanic Black and Hispanic patients** and with adverse social drivers (lower education, poverty, insurance status, unmet social needs), **independent of incontinence severity.**
- **Oral pharmacotherapy can cost hundreds of dollars per month** in the United States; **no guideline-directed drug is universally covered at low cost,** even where generics exist.
- **Cost-effectiveness analysis (US payer perspective, versus conservative care): sacral nerve stimulation (SNS) was the costliest option over 10 years; intravesical onabotulinumtoxinA 100 U was the most cost-effective.**
- **Societal cost of OAB in the United States runs to tens of billions of dollars a year and is rising;** per capita costs are similar by sex, with **higher direct non-medical costs in women** (more incontinence).

**Social Drivers of Health**

- **Lower socioeconomic status is associated with UUI, more strongly as severity increases.**

- **Unmet social needs associated with OAB ± UUI:** food insecurity, financial strain, employment difficulty, access to care (transport and cost), housing instability, **lack of working toilets, plumbing and utilities**, poor social support, interpersonal stress and violence.
- **Implication:** patient-centred screening for unmet social needs belongs in comprehensive OAB care.

**Concept — OAB is common, under-presented and under-treated:** *the burden is driven by UUI and by social context, while the modest efficacy and poor persistence of anticholinergics explain why so few symptomatic people end up on effective treatment.*

## PATHOPHYSIOLOGY AND ETIOLOGY

- **Pathophysiology is multifactorial and spans every level of neural control.**
- **DO is no longer the sine qua non of OAB** — definition now rests on urgency, and DO has limited clinical utility.
- **Only about half of patients with DO have urgency;** only 44–69% of patients with urgency show DO on cystometry.
- **No animal model of OAB exists** — animals cannot report urgency — so research uses **surrogate markers** in three areas: **sensory activity, motor control and LUT reflexes.**
- **Sensory (afferent):** urgency is presumed afferent-derived. Stages: **signal transduction → afferent traffic → gating → sensitisation → conscious perception.** The **urothelium and suburothelium** drive transduction via mediators, cell interactions, cytokines and growth factors; **gut inflammation can sensitise bladder afferents;** functional brain imaging is beginning to map conscious perception.
- **Motor:** detrusor motility and its drivers (efferent nerves, interstitial cells, local mediators) summate into **whole-bladder contractility**, which determines urodynamic findings. **Localised micromotions sit beside quiescent areas;** intravesical pressure will not be understood until both are.
- **Reflex integration:** afferents are integrated at several central nervous system (CNS) levels (table below); **the periaqueductal grey and pontine micturition centre (PMC) in midbrain and brainstem are the main regulatory level,** and **reflex sacral voiding matters in neonates and neurological disease.**
- **Functional brain imaging** implicates the **insula and prefrontal cortex;** **altered regional activity** is reported in OAB, and brain activity changes with **SNS, tibial nerve stimulation and oral anticholinergics.**

FIGURE 110.4 LEGEND — LEVELS OF NEURAL CONTROL OF THE LUT

| Level                          | Afferent                                        | Integration                                                        | Efferent                                                                          |
|--------------------------------|-------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Forebrain                      | Conscious awareness (sensation)                 | Homeostasis, <b>volition</b> , emotion, social suitability         | <b>Inhibition for storage; disinhibition for voiding</b>                          |
| Midbrain / brainstem (PMC)     | Subconscious monitoring of filling              | <b>Voiding reflex</b> , homeostasis (input from vegetative organs) | Bladder inhibition / outlet excitation for storage; <b>bladder–outlet synergy</b> |
| Sacral spinal cord             | Subconscious monitoring of filling              | <b>Immature voiding reflexes</b> (infants)                         | Bladder and outlet motor nuclei                                                   |
| Pelvic plexus / LUT (cellular) | Urothelium, interstitial cells, local mediators | Myovesical plexus — <b>micromotions</b>                            | Widespread bladder contraction                                                    |

- **Descending forebrain input to the PMC switches storage to voiding,** with synergic regulation of the sacral nuclei (detrusor contraction, outlet relaxation).

### Afferent Mechanisms in Overactive Bladder

- **Premise:** increased sensory activity → increased sensation → OAB. Afferent endings are **densest in the suburothelial connective tissue.**

**BLADDER AFFERENTS — THE PAIRED CONTRAST**

|                                            | <b>Aδ fibres</b>                                                          | <b>C fibres</b>                                                                                                |
|--------------------------------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Calibre and myelination                    | <b>Small calibre, thinly myelinated</b>                                   | <b>Smallest calibre, unmyelinated</b>                                                                          |
| Normal role                                | <b>Predominant afferent; senses volume and detrusor contractile state</b> | <b>"Silent"</b> — respond only to high-intensity stimuli (extreme distension, cold, heat, chemical irritation) |
| In pathological states (animal OAB models) | —                                                                         | <b>Become spontaneously active and hypersensitive to low-intensity input</b> , via spinal second-order neurons |

- **Two broad categories: increased afferent signalling from the bladder or decreased CNS capacity to modulate afferent signals.**

**Mechanisms of Increased Afferent Activity****UROTHELIUM-BASED VERSUS MYOGENIC HYPOTHESIS**

|                     | <b>Urothelium-based</b>                                                                                            | <b>Myogenic</b>                                                                                                                                                           |
|---------------------|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary site        | <b>Urothelium / suburothelium</b>                                                                                  | <b>Detrusor smooth muscle</b>                                                                                                                                             |
| Trigger             | Local <b>mechanical, osmotic, inflammatory, chemical</b> stimuli alter receptor/channel expression and sensitivity | <b>Increased spontaneous excitation</b> within muscle                                                                                                                     |
| Transmission        | <b>Mediator release acts on adjacent afferents</b> — urothelium as transducer                                      | <b>Enhanced propagation</b> of excitation across an excessive proportion of the wall                                                                                      |
| Supporting findings | Intrinsic homeostatic aberration, possibly to unidentified local stimuli                                           | <b>Patchy denervation</b> (any cause of DO) → <b>receptor upregulation, altered membrane potential</b> → spontaneous contraction; <b>structural changes</b> favour spread |
| Result              | Excess afferent traffic adds to fullness signal → <b>urgency, micturition reflex</b>                               | Exaggerated <b>micromotions</b> → urgency; <b>DO when coordinated</b> across much of the bladder                                                                          |

- **Micromotions:** localised contractions, first described in pigs, present in **normal** animals and humans.
- **Normally:** detected by afferents as **filling sensation without any change in detrusor pressure**.
- **In OAB and DO models:** **exaggerated** → urgency; **coordination across a large area** → urodynamic DO.

*Other mechanisms of increased afferent signalling.* **Signals arising outside the bladder can act through shared afferent pathways** — convergence at **dorsal root ganglia, spinal cord or brain** gives **pelvic organ crosstalk**. - **Bowel is the best-understood partner** — it shares thoracic and sacral spinal sensory innervation. - **Animal work:** rectal distension or inflammation causes detrusor inhibition and/or overactivity via **convergent bladder-bowel C-fibre afferents**; colonic stimulation heightens bladder pain responses. - **Clinical correlates:** rectal distension changes bladder capacity, sensation and DO; constipation and straining impair emptying and worsen storage and voiding symptoms.

*Abnormal handling of afferent signals.* Continuous normal afferent traffic ("**afferent noise**") conveys pain, mechanical, chemical and motor information; **only a fraction reaches sensation**, the rest serves reflexes. - **The CNS suppresses unnecessary sensations and facilitates homeostatic ones.** - **If disease or dysfunction impairs this modulation, normally unconscious signals are perceived as urgency.**

**Other Pathophysiologic Mechanisms**

- **Ischaemia-reperfusion** (e.g. atherosclerosis, arterial obstruction) → **hypoxia and oxidative stress** → damage to urothelium, smooth muscle, microvasculature and nerves.

- **Chronic inflammation:** raised **cytokines, chemokines and growth factors** in bladder tissue, urine and blood.
- **Urinary microbiome:** composition **differs in UUI and OAB**; its link to bladder–brain signalling is unknown.

**Concept** — *every mechanism converges on one final common pathway — more afferent signal reaching awareness: either the bladder (urothelium, muscle), a neighbouring organ or ischaemia–inflammation turns the signal up, or the CNS fails to turn it down.*

### Etiology and Attributable Risk Factors

- **Cause usually unknown and multifactorial**; risk factors initiate or amplify the mechanisms above.
- **Ageing:** clearly associated, through widespread cellular and CNS change.
- **Sex:** higher prevalence in **women, especially younger women**, and more UUI.
- **Bladder outlet obstruction (BOO):** the standard **animal model of DO**, but the **human link is not clear-cut**:
- **OAB prevalence in women (BOO rare) is broadly similar to that in men (BOO common).**
- **DO often persists after prostate surgery.**
- **Conclusion:** BOO is relevant but probably **indirect** — **accelerating age-related change** rather than directly causal.
- **Race and ethnicity:** associations exist; mechanisms unknown.
- **Psychological factors:** **depression and anxiety** most studied; also self-esteem, intimacy, embarrassment and shame.
- **OAB wet carries more psychological burden than OAB dry.**
- Possible mechanisms: **serotonergic CNS pathways; hypothalamic–pituitary–adrenal axis dysregulation** — **stress acts via this route.**
- **Women with OAB and at least mild anxiety** have worse OAB QoL and **greater temporal summation of heat pain** — a marker of central sensitisation.
- **Toileting behaviours:** women with OAB show **convenience ("just in case") voiding, delayed voiding, straining and positional preferences** — possible risk factors.
- **Coping behaviours:** **strategising** (mapping toilets, altering plans), **fluid restriction** and unhealthy toileting — associated with **stress and anxiety independent of symptoms**, self-reinforcing, and can sustain a **negative perception of the bladder even after objective improvement.**
- **Wider associations:** bowel function, metabolic syndrome, hormonal status, psychosocial burden.

#### KEY POINTS: PATHOPHYSIOLOGY AND ETIOLOGY

- Main integration in midbrain and brainstem; failed higher modulation → unconscious signals felt as urgency.
- Two bladder-centric theories: urothelium-based and myogenic.
- BOO and OAB: separate consequences of ageing, not directly causal.

**Concept** — **BOO is a model, not a cause:** *similar OAB prevalence in women and men and persistence of DO after de-obstruction show that ageing, psychology and behaviour sit upstream of, or alongside, obstruction.*

## CLINICAL ASSESSMENT

### Initial Evaluation

- **OAB is diagnosed by characteristic LUTS; empirical treatment** may start on history and non-invasive tests that characterise symptoms and exclude an acute cause.

### GUIDELINE BODIES — MINIMUM INITIAL EVALUATION

|                                                   | AUA/SUFU (American Urological Association / Society of Urodynamics, Female Pelvic Medicine & Urogenital Reconstruction), non-neurogenic OAB | EAU (European Association of Urology), male and female non-neurogenic LUTS   |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Minimum                                           | <b>History, physical examination, urinalysis</b>                                                                                            | <b>Concurs</b>                                                               |
| Questionnaires                                    | Recommends UDI / UDI-6, IIQ or OAB-q                                                                                                        | <b>Strongly encourages validated questionnaires</b>                          |
| Bladder diary                                     | Not specified                                                                                                                               | <b>Consider bladder diary</b>                                                |
| Urodynamics, cystoscopy, renal/bladder ultrasound | <b>Not in initial work-up of uncomplicated patient</b>                                                                                      | Urodynamics only when the result will change management (consensus position) |

### History

Elements to cover: - **Presence and bother of each OAB symptom** (urgency, UII, daytime frequency, nocturia), **other storage symptoms (SUI)** and **voiding symptoms suggesting BOO** (straining, hesitancy, incomplete emptying); also **post-micturition symptoms, dysuria, haematuria**. - **Questionnaires help but are limited by poor health literacy or language barriers** (Table 110.3). - **Fluid intake — type and volume:** caffeine and polydipsia worsen LUTS; patients often already restrict intake. - **Occult neurological disease:** e.g. **new OAB with erectile dysfunction or tremor**, or previous **spinal surgery / disc disease** ± sensory or musculoskeletal symptoms. - **Obstetric, gynaecological, surgical and radiotherapy history; bowel symptoms; medications.** - **Faecal incontinence:** 7–15% of the general population, but close to 50% of women with OAB. - **Comorbidity** that (a) causes LUTS directly (e.g. **polyuria from poorly controlled diabetes**), (b) constrains drug choice, or (c) increases urine output (**obstructive sleep apnoea, psychogenic polydipsia, lithium-induced diabetes insipidus**).

### RELATIVE CONTRAINDICATIONS IDENTIFIED AT HISTORY

| Drug class             | Relative contraindications                                                                                             |
|------------------------|------------------------------------------------------------------------------------------------------------------------|
| Antimuscarinics        | Poorly controlled closed-angle glaucoma; cognitive impairment; history of urinary retention; impaired gastric emptying |
| β3-adrenergic agonists | Poorly controlled hypertension; cardiac arrhythmia                                                                     |

**TABLE 110.3 DESCRIPTION OF COMMON PATIENT-REPORTED OUTCOME MEASURES FOR OVERACTIVE BLADDER, URGE INCONTINENCE, AND LOWER URINARY TRACT SYMPTOMS**

| Group                            | Instrument (items)            | Measures                                                   |
|----------------------------------|-------------------------------|------------------------------------------------------------|
| ICIQ modules — OAB               | ICIQ-OAB (4)                  | Frequency, nocturia, urgency, UII                          |
|                                  | ICIQ-OABqol (26)              | OAB QoL — coping, worry, sleep, social interaction         |
| ICIQ modules — sex-specific LUTS | ICIQ-FLUTS (12; long form 18) | Female filling, voiding, incontinence and bother           |
|                                  | ICIQ-MLUTS (13; long form 23) | Male voiding, incontinence, frequency, nocturia and bother |

| Group                                  | Instrument (items)                                | Measures                                                                               |
|----------------------------------------|---------------------------------------------------|----------------------------------------------------------------------------------------|
| ICIQ modules —<br>incontinence and QoL | ICIQ-UI Short Form (4)                            | Incontinence frequency, amount, perceived cause                                        |
|                                        | ICIQ-LUTSqol (22); I-QOL / ICIQ-Uqol (22)         | LUTS QoL (life restriction, emotional burden, preventive measures); incontinence QoL   |
| OAB-specific, non-ICIQ                 | OAB-q Short Form (19)                             | OAB QoL with and without incontinence                                                  |
|                                        | OAB-SS, OAB Symptom Score (7)                     | Symptom severity, both sexes — frequency, nocturia, urgency, UII                       |
|                                        | OAB-V8, OAB Awareness Tool (8)                    | <b>Primary-care screening</b>                                                          |
| Both sexes, full LUTS spectrum         | LURN SI-29 (27 + 2 sex-specific); LURN SI-10 (10) | Urgency, incontinence, voiding difficulty, nocturia, pain, frequency, post-micturition |
|                                        | LUTSS (14); LIS (21); LUSQ (10)                   | Storage and voiding with bother; LUTS impact on QoL; presence and severity of symptoms |
| Female-specific                        | UDI-6 (6); B-SAQ (8); ISS (8); UISS (10)          | LUTS and incontinence; female LUTS; severity; severity and impact                      |
|                                        | QUID (6)                                          | <b>Distinguishes type of incontinence</b>                                              |
|                                        | PRAFAB (5)                                        | <b>Treatment effect</b> on incontinence                                                |
| Incontinence impact                    | IIQ-7 (7)                                         | Impact of incontinence on QoL                                                          |

Abbreviations: ICIQ, International Consultation on Incontinence Questionnaire; LURN SI, Lower Urinary Tract Research Network Symptom Index; UDI, Urogenital Distress Inventory; IIQ, Incontinence Impact Questionnaire; QUID, Questionnaire for Urinary Incontinence Diagnosis; PRAFAB, Protection, Amount, Frequency, Adjustment, Body image; B-SAQ, Bladder Self-Assessment Questionnaire; ISS, Incontinence Symptom Severity Index; UISS, Urinary Incontinence Severity Score; LUTSS, Lower Urinary Tract Symptom Score; LIS, Leicester Impact Scale; LUSQ, Leicester Urinary Symptom Questionnaire.

- **What the table is for:** matching the instrument to the question — **symptom, bother, QoL, screening, type-of-incontinence or treatment response.**
- **Pattern:** ICIQ modules pair a symptom module with a QoL module; only LURN, LUTSS, LIS and OAB-SS cover both sexes; most others are **female-only**.
- **Trap:** OAB-V8 is a screening tool, QUID distinguishes incontinence type, PRAFAB measures treatment effect — none is a severity score for OAB.

## Additional Assessment

### Physical Examination

- **Directed examination is mandatory even when the diagnosis seems straightforward:** abdomen and pelvis, extremities, basic neurological examination, **cognition and frailty**.
- **General appearance** (gait, mobility, stature, frailty) is a surrogate for performance status and neurological disease.
- **Abdomen:** scars, masses, hernias, tenderness; **suprapubic tenderness or distension**.
- **Lower limbs:** peripheral oedema — fluid shifts on lying down, **relevant to nocturia**.
- **Pelvic and genital examination** ( $\pm$  digital rectal examination, DRE, in men) is perhaps the most important component.
- **Women:** pelvic floor tone and pain, **pelvic organ prolapse, urethral prolapse or caruncle, labial adhesions, vaginal oestrogen status** (atrophic vaginitis contributes to OAB), **supine stress test** for SUI when indicated.

- **Men:** prostate size and abnormality on DRE.
- **All:** perineal sensation and anal tone.
- **Cognition:** brief assessment in all (observation at interview); if concerned, **Mini-Mental State Examination (MMSE)**.
- **Frailty:** consider formal assessment in older patients — **frailty predicts poorer surgical outcomes and complications**, including in urology.
- **Timed Up and Go Test (TUGT):** stand from sitting → walk 10 feet, timed; simple, usable in any clinic.

### Postvoid Residual Volume

- **Not required at initial work-up, nor before lifestyle, behavioural or oral drug treatment, in uncomplicated patients.**
- **Measure postvoid residual (PVR) in:** complicated patients; **risk factors for retention; obstructive or voiding symptoms** (especially men); **previous anti-incontinence or prostate surgery; neurological disease.**
- **Method:** portable ultrasound bladder scanner (non-invasive, cheap, comfortable) → **catheterisation** if indicated or no scanner. Palpation may reveal a distended bladder.

### Urinalysis

- **Mandatory in all** — excludes **UTI, haematuria, pyuria.**
- **Culture if infection suspected; microhaematuria → full haematuria evaluation.**

### Other Laboratory Testing

- **Serum creatinine and urea:** not routine, but review if available.
- **Prostate-specific antigen (PSA):** consider in male LUTS if it would alter treatment or decisions.
- **Urine culture at discretion in acute-onset irritative symptoms with a negative dipstick** — **urinalysis is unreliable at low bacterial counts.**

### Bladder Diary

- **Record of fluid intake and voiding behaviour.** Most useful when patients **cannot describe or recall** their pattern, or when **polyuria** is suspected.
- **Uses:** diagnosis; tailoring **behavioural and lifestyle change; objective measure of treatment response; self-monitoring effect** — patients recognise triggers.
- **Duration:** ideally 3–7 days; **because this is burdensome, a minimum of 24–48 hours.**
- **Records:** time and **volume** of each void, incontinence episodes and circumstances, **degree of urgency** (Table 110.4).
- **OAB pattern:** frequent, small-volume voids.
- **Daytime frequency counts waking voids, including the last before sleep and the first on rising.**
- **Maximum voided volume = functional bladder capacity** — a marker of OAB severity and of DO severity.
- **Limitation:** compliance; **electronic diaries and smartphone applications** help.

TABLE 110.4 BLADDER SENSATION SCALE USED IN THE INTERNATIONAL CONSULTATION ON INCONTINENCE QUESTIONNAIRE BLADDER DIARY

| Grade | Sensation                                                                           |
|-------|-------------------------------------------------------------------------------------|
| 0     | No desire — voided for "social reasons" (before going out, uncertain toilet access) |
| 1     | Normal desire, no urgency                                                           |
| 2     | Urgency that passed before reaching the toilet                                      |

| Grade | Sensation                                              |
|-------|--------------------------------------------------------|
| 3     | <b>Urgency persisting to the toilet, no leakage</b>    |
| 4     | Urgency with <b>leakage before reaching the toilet</b> |

From Bright E, Cotterill N, Drake M, et al.: Developing a validated urinary diary: phase 1, Neurourol Urodyn 31(5):625–633, 2012.

- **What the table is for:** grading each void so that **urgency, not just frequency**, is captured on the diary.
- **Pattern:** grades 0–1 are **normal or pre-emptive** voids; **grade 2 upward is urgency**; the top grade is UUI.
- **Trap:** a grade 0 ("just in case") void inflates frequency without urgency — **convenience voiding**, a behaviour, not OAB.

### Patient-Reported Outcome Measures and Symptom Questionnaires

- **OAB is symptom-based**, so **patient-reported outcome measures (PROMs)** of frequency, severity and bother are central to evaluation and to **monitoring response**.
- **International Consultation on Incontinence (ICI)** grades PROMs A to C, with "+" where further content-validity evidence exists.
- **ICIQ modules** (first developed after the 1996 ICI): standardised, easy to use; **pair each symptom module with its HRQoL module**.
- **OAB set:** ICIQ-MLUTS / ICIQ-FLUTS (general, sex-specific) + ICIQ-OAB (condition-specific) + ICIQ-OABqol (QoL); ICIQ-UI core incontinence module. Many exist in short and long forms.
- **AUA/SUFU recommend UDI, UDI-6, IIQ or OAB-q.**
- **LURN SI:** one instrument for **both sexes and the full LUTS spectrum** (OAB, pelvic floor disorders, all incontinence types); **29-item long form (with two sex-specific items) and 10-item short form**.

### Advanced Evaluation

- **Urodynamics, cystoscopy and renal/bladder ultrasound are not for the initial work-up of the uncomplicated patient.**

### URODYNAMICS — WHEN TO USE, WHEN NOT

| Indicated                                                                                                                                               | Not indicated                                                                                                                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Result (compliance, DO, other abnormality) <b>will change the decision to pursue or avoid an invasive, irreversible or potentially morbid treatment</b> | <b>Initial work-up</b> of uncomplicated OAB                                                                                      |
| <b>Refractory to multiple OAB treatments</b>                                                                                                            | <b>Index patient</b> — woman, low PVR, no significant prolapse, no neurological disease, minimal SUI — no proven outcome benefit |
| <b>Suspected BOO</b> , e.g. persistent or new OAB <b>after an outlet procedure</b>                                                                      |                                                                                                                                  |

- **Urodynamics may improve treatment satisfaction** but has not been shown to improve outcomes in the index patient.
- **Aim:** **reproduce the patient's symptoms** and identify factors that change management.
- **Two urodynamic diagnoses associated with OAB:** **DO** and **early and/or exaggerated filling sensation** ("increased bladder sensation" — **early and persistent desire to void**, the usual finding when DO is absent).
- **Antimuscarinics before the study** — **either way is defensible:**

| Stop the drug                   | Continue the drug                                                |
|---------------------------------|------------------------------------------------------------------|
| Best chance of demonstrating DO | Shows the mechanism of residual refractory symptoms on treatment |

- DO correlates with urgency severity and with incontinence:

#### DO ON CYSTOMETRY BY SYMPTOM GROUP

| Group         | Men    | Women                                                        |
|---------------|--------|--------------------------------------------------------------|
| OAB dry       | 60–70% | 40–50%                                                       |
| OAB wet (UII) | >90%   | Absent in 30–40% — UII is a less reliable predictor in women |

- Why women's UII predicts DO less well: SUI misread as urgency leakage, and pre-emptive voiding for SUI misread as frequency or urgency.

#### FIGURE 110.5 LEGEND

- Overlap of symptom-based OAB (increased daytime frequency, urgency, nocturia, UII) with urodynamic DO: correlation best for UII, worst for nocturia.
- Technique points:
- Provocation if DO not seen: whatever triggers symptoms — running water, standing.
- Filling rate: fast filling can induce DO, and repeated DO can obscure other parameters.
- Position: seated or standing — OAB is experienced upright. If severe low-volume DO masks SUI, a second fill supine usually reduces DO.
- Always assess for BOO or inefficient emptying, for two reasons:
- Storage-directed treatments may impair voiding → intermittent self-catheterisation. Each 10 mL rise in PVR raises the risk of catheterisation after onabotulinumtoxinA by 27%.
- Severe BOO is associated with DO, so OAB may not improve unless BOO is treated — yet relieving BOO does not guarantee improvement: DO persists in up to 60% of men after transurethral resection of the prostate (TURP).

#### KEY POINTS: OVERACTIVE BLADDER CLINICAL ASSESSMENT

- Minimum: history, examination, urinalysis; consider malignancy, neurological or systemic disease, raised PVR.
- Culture and PVR as indicated; diaries and PROMs clarify and track.
- No urodynamics, cystoscopy or ultrasound initially; urodynamics when it changes an invasive decision or after multiple failures.

**Concept** — *assessment escalates only when the answer would change management: the uncomplicated patient is treated on history, examination and urinalysis, and every further test (PVR, diary, urodynamics) is justified by a specific decision it informs.*

## MANAGEMENT

- OAB is chronic, affects QoL and is not life-threatening — counsel in that frame.
- No treatment is an acceptable choice after counselling and exclusion of other conditions — best where costs (financial, physical, opportunity) outweigh benefit, or the patient cannot perceive improvement (significant cognitive impairment).
- Consider contributors such as limited mobility — curing the bladder will not cure immobility-driven incontinence.

- **Exception — incontinence-associated dermatitis:** painful, risks infection, pressure ulcers and wounds → may justify more aggressive treatment.
- **Set individual goals** (fewer pads, dryness, activities, independence, avoiding lifelong drugs or surgery); goals may change.
- **Realistic expectations:** most treatments **improve but do not eliminate** symptoms; improvement is gradual and irregular. **Critical before invasive treatment.**
- **Education** on normal bladder function empowers patients; **success depends on an engaged patient.** The **consistent placebo response in OAB trials** shows the value of non-pharmacological input, much of it education.
- **AUA/SUFU "lines" (first to fourth) rank invasiveness, adverse-event severity and reversibility against benefit — they are not a mandatory linear path; lines are often combined.**

### Conservative Management

- **Options:** bladder training, bladder control strategies, **pelvic floor muscle therapy (PFMT)**, **weight loss**, **fluid management.** Evidence is mainly on **incontinence in women.** Goal is improvement, not cure.
- **Fluid reduction:** improves symptom scores at the cost of **thirst and under-hydration**; many patients **already restrict** as coping. **A 25% reduction in intake meaningfully reduces symptoms** — encourage where appropriate.
- **Caffeine reduction:** improves urgency and frequency but not consistently incontinence — patient decides if the trade-off is worth it.
- **Weight loss:** reduces SUI and UUI. In a randomised trial of obese incontinent women, **intensive weight loss (≈8 kg vs ≈1.5 kg in controls) meaningfully reduced UUI episodes.** **Bariatric surgery: each 5% weight loss → ≈7–8% lower risk of incontinence** — discuss honestly in severe obesity.
- **PFMT and bladder training: 50–80% reduction in incontinence episodes**, negligible risk, **endorsed by multiple societies.**
- **Combine behavioural with drug therapy** where the patient wishes — limited evidence of better outcomes.

### Pharmacologic Management

- **Two classes for OAB:** antimuscarinics and **β3-adrenergic agonists.**
- **Counsel on antimuscarinic side effects:** dry mouth, constipation, **cognitive effects**, visual impairment.
- **Dementia:** Society of Urodynamics, Female Pelvic Medicine & Urogenital Reconstruction (SUFU) white paper — **chronic use (>3 months) → 10–65% increased risk of new-onset dementia.**
- → **Shared decision-making** with patient and/or caregivers.
- → **Preferred:** **β3 agonists**, or **extended-release trospium, darifenacin or fesoterodine** (properties that limit CNS effects).
- **Response is idiosyncratic** — expect adjustment of agent and dose.
- **Antimuscarinics raise the likelihood of cure over placebo by 20–70%, but withdrawal (inefficacy or adverse effects) is high.**
- **Extended-release over immediate-release** — less dry mouth, fewer cognitive effects.
- **Inadequate response or intolerance → change dose or switch agent; then different agent or combination**, which may avoid invasive investigation and treatment.

**DRUGS FOR OAB AND MALE STORAGE LUTS**

| Class                                        | Role                                       | Preferred agents / key point                                                                             | Signature risk                                                                                                            |
|----------------------------------------------|--------------------------------------------|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>Antimuscarinics</b> (oral or transdermal) | OAB ± UUI; add-on in men with storage LUTS | <b>Extended-release; trospium, darifenacin, fesoterodine</b> limit CNS effects                           | <b>Dementia with &gt;3 months' use</b> ; dry mouth, constipation, visual blurring; <b>acute retention up to 3% in men</b> |
| <b>β3-adrenergic agonists</b>                | OAB; add-on in men with storage LUTS       | <b>Preferred class</b> where CNS risk matters                                                            | Relative contraindications as above                                                                                       |
| <b>α1-adrenergic antagonists</b>             | <b>First-line for male voiding LUTS</b>    | <b>Combination with antimuscarinic beats either alone</b> in benign prostatic hyperplasia (BPH) with OAB | Retention risk rises when antimuscarinic added                                                                            |

- **Men:** combination therapy's retention risk → **caution when PVR is elevated.**

**Procedural or Surgical Management**

- **"Third-line" (AUA/SUFU):** after failure of, or unsuitability for, conservative and drug therapy — or **earlier** when medical, social or personal barriers make those unworkable.
- **More invasive, more adverse events.**
- **Frail and elderly:** frail patients **do respond well**, but **frailty predicts functional decline, discharge to skilled or assisted nursing, and 1-year morbidity up to 50% even after minor urological procedures** — weigh carefully.
- **Choice (including no further treatment) is by patient preference after informed discussion.**

**THIRD-LINE THERAPIES FOR REFRACTORY OAB**

|                    | <b>Percutaneous tibial nerve stimulation (PTNS)</b>                                                     | <b>Intradetrusor onabotulinumtoxinA</b>                                                                                                          | <b>Sacral nerve stimulation (SNS)</b>                                                                                    |
|--------------------|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Efficacy           | <b>Clinical response 60–80%</b> of drug-refractory patients                                             | Frequency ↓ 29%, urgency ↓ 38%, <b>incontinence episodes ↓ 59%</b> (pooled placebo-controlled trials); <b>≈25% completely dry</b>                | <b>5-year success ≈70–80%</b> (fewer UUI episodes and voids)                                                             |
| Main drawback      | <b>Weekly office sessions for 12-week induction; response delayed to weeks 8–10</b> — limits compliance | <b>Symptomatic UTI ≈20%; intermittent catheterisation 0–12% in trials, but retention ≈20% in women and up to 35% in men in real-world series</b> | <b>Revision &gt;30% at 5 years</b> (stimulation change, pain, inefficacy); <b>battery replacement eventually for all</b> |
| Risk profile       | <b>Lowest</b>                                                                                           | Moderate                                                                                                                                         | Highest; <b>costliest</b>                                                                                                |
| Cost-effectiveness | —                                                                                                       | <b>Most cost-effective (100 U)</b>                                                                                                               | <b>Costliest</b>                                                                                                         |
| Recent advances    | —                                                                                                       | —                                                                                                                                                | <b>Rechargeable, longer battery life, MRI-compatible devices</b> widen eligibility                                       |

- **What this table is for:** the trade-off between **burden of attendance (PTNS), risk of retention and UTI (onabotulinumtoxinA) and revision and battery surgery (SNS).**
- **Trap:** trial catheterisation rates for onabotulinumtoxinA **understate real-world retention**, especially in men.

**Augmentation Cystoplasty or Urinary Diversion**

- **Rarely,** OAB is severe enough to justify **augmentation cystoplasty or urinary diversion.**

- **Evidence is mostly neurogenic** — QoL impact in non-neurogenic OAB is poorly known.
- **Weigh against short- and long-term risks of major surgery and against containment (pads) or indwelling catheter.**

#### KEY POINTS: CLINICAL MANAGEMENT

- **Conservative** → antimuscarinic (oral or transdermal) or  $\beta_3$  agonist → tibial or sacral neuromodulation or onabotulinumtoxinA → augmentation or diversion, by shared decision-making.
- **Chronic antimuscarinics carry dementia risk.**

### Approach

- **Sixth ICI** provides evidence-graded pathways for **women, men and the frail elderly** (Figure 110.6); **AUA/SUFU** and **SUFU OAB Clinical Care Pathway** provide algorithms.
- **Common sequence:** careful assessment to confirm OAB and exclude alternatives → **patient education** → **conservative** (lifestyle, behavioural therapy, PFMT) → **pharmacological** → **invasive** (tibial nerve stimulation, sacral neuromodulation, chemodenervation).
- **At every step reassess symptoms, side effects and preference.**
- **Minimum trial before judging efficacy:** 4–12 weeks for behavioural therapy and 4–12 weeks for drugs.

**FIGURE 110.6 LEGEND — ICI ASSESSMENT AND MANAGEMENT OF OAB IN MEN, WOMEN AND THE FRAIL ELDERLY**

| Stage              | Men                                                                                                                                                                                                                                                                          | Women                                                                                                                                                      | Frail elderly                                                                                                                                                                                                                                                       |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Presentation       | Urgency/frequency ± incontinence                                                                                                                                                                                                                                             | Incontinence with mixed symptoms                                                                                                                           | <b>Active case finding</b>                                                                                                                                                                                                                                          |
| Initial assessment | General: symptom score (incl. <b>frequency-volume chart</b> , FVC) and questionnaire, QoL and desire for treatment, <b>urinalysis ± culture</b> (treat infection, reassess). Examination: abdominal, <b>rectal, sacral, neurological</b> ; pelvic floor function; <b>PVR</b> | General as for men. Examination: abdominal, <b>pelvic, perineal; cough test for SUI; oestrogen status</b> ; voluntary pelvic floor contraction; <b>PVR</b> | Assess and treat <b>potentially treatable conditions and comorbidity</b> ; QoL, goals, <b>patient and caregiver preference</b> ; targeted examination incl. <b>cognition</b> , neurological, rectal; urinalysis; <b>FVC or wet checks, especially with nocturia</b> |
| Presumed diagnosis | OAB ± UUI                                                                                                                                                                                                                                                                    | <b>Mixed incontinence — treat most bothersome symptom first</b>                                                                                            | —                                                                                                                                                                                                                                                                   |
| Initial management | Lifestyle; PFMT ± biofeedback; scheduled voiding; containment products; <b>antimuscarinic, + <math>\alpha</math>-blocker if BOO</b>                                                                                                                                          | Lifestyle; PFMT for SUI or OAB; bladder retraining for OAB; antimuscarinic                                                                                 | Lifestyle; behavioural therapy; <b>consider trial of antimuscarinic; treat significant PVR</b>                                                                                                                                                                      |

| Stage      | Men                                                                                                                                                                                                                                                                                                                                                                  | Women                                                                                                                                                                                                                                                                                                                                                   | Frail elderly                                                                                                                                                  |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| On failure | <b>Specialist:</b> urodynamics, urinary tract imaging, urethrocystoscopy if indicated → mixed incontinence or <b>DO incontinence</b> → <b>treat major component first:</b> $\alpha$ -blocker, <b>5<math>\alpha</math>-reductase inhibitor (5ARI)</b> , correct BOO, antimuscarinic → if fails: <b>neuromodulation; intermittent catheterisation + antimuscarinic</b> | <b>Specialist:</b> assess <b>prolapse and urethral mobility</b> , imaging, urodynamics → mixed ( <b>urodynamic stress incontinence</b> , USI, / DO incontinence, DOI) or DOI → treat most bothersome first → <b>SUI surgery</b> (bulking agents, tapes and slings, colposuspension) or <b>onabotulinumtoxinA, neuromodulation, bladder augmentation</b> | <b>Reassess for treatable comorbidity and functional impairment</b> → if still insufficient or severe symptoms: <b>referral per preference and comorbidity</b> |

Derived from the algorithms of the fifth International Consultation on Incontinence (Abrams et al., 2013).

- **Pattern:** men — treat the outlet alongside the bladder; women — treat the most bothersome component of mixed incontinence first; frail elderly — treat comorbidity and function first, then the bladder.

**Concept** — *management is a ladder ordered by invasiveness and reversibility, not a compulsory sequence: each rung buys greater efficacy at greater risk, so patient goals, frailty and PVR — not the next line on the algorithm — decide how far and how fast to climb.*